

# Métodos Matemáticos de la Física II

## Cheat Sheet: Solving PDEs in Mathematical Physics

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### I. THE GENERAL PHILOSOPHY

Most linear PDEs encountered in physics can be written schematically as

$$(\mathcal{M}u)(\mathbf{x}, t) = (\mathcal{L}u)(\mathbf{x}, t) + f(\mathbf{x}, t),$$

where:

- $\mathcal{L}$  is a spatial differential operator,
- $\mathcal{M}$  is a temporal differential operator,
- $f$  is a source term.

The central idea is:

Geometry  $\rightarrow$  Boundary Conditions  $\rightarrow$  Spatial Eigenvalue Problem  $\rightarrow$  Eigenfunctions  $\rightarrow$  Expansion of Solutions

The fundamental step is therefore not solving the PDE directly, but solving the associated spatial eigenvalue problem.

### II. SPATIAL EIGENVALUE PROBLEM

Given a spatial operator  $\mathcal{L}$ , solve the equation

$$\mathcal{L}\phi_n = -\lambda_n\phi_n$$

subject to the prescribed boundary conditions. This reduces to the Helmholtz equation in the particular case where  $\mathcal{L} = \nabla^2$ .

The eigenfunctions  $\phi_n$ :

- form an orthogonal basis,
- determine the structure of the solution, and
- are reused for Laplace, Poisson, Heat, Wave, Helmholtz and Schrödinger equations.

### III. STURM-LIOUVILLE THEORY

Most one-dimensional eigenvalue problems can be written as a Sturm-Liouville problem, which is about solving the ODE

$$-\frac{d}{dx} \left( p \frac{dy}{dx} \right) + qy = \lambda w y.$$

under prescribed boundary conditions, and where  $p : x \rightarrow p(x) \in \mathbb{R}$ ,  $q : x \rightarrow q(x) \in \mathbb{R}$  and  $w : x \rightarrow w(x) \in \mathbb{R}$  denote some prescribed functions and  $y : x \rightarrow y(x) \in \mathbb{R}$  denotes a solution.

Properties:

- **Real eigenvalues:**

$$\lambda_n \in \mathbb{R}.$$

- **Orthogonality:** Two eigenfunctions  $y_n$  and  $y_m$  obtained for different eigenvalues  $\lambda_n$  and  $\lambda_m$  (i.e. for  $n \neq m$ ) satisfy the orthogonality condition

$$\langle y_n, y_m \rangle = 0$$

where

$$\langle f, g \rangle = \int_a^b f^*(x)g(x)w(x) dx$$

denotes an inner product.

- **Completeness:** Any sufficiently regular function can be expanded as

$$f(x) = \sum_n a_n y_n(x)$$

where

$$a_n = \frac{\langle y_n, f \rangle}{\langle y_n, y_n \rangle}$$

#### IV. BOUNDARY CONDITIONS

Assume that  $\Omega$  is the spatial domain under consideration.

##### A. Dirichlet

Value prescribed:

$$u(\mathbf{x}, t) = g(\mathbf{x}, t) \quad \text{for all } \mathbf{x} \in \partial\Omega \text{ and all } t.$$

Examples:

- fixed temperature,
- fixed electrostatic potential.

##### B. Neumann

Normal derivative prescribed:

$$\frac{\partial u}{\partial \hat{n}}(\mathbf{x}, t) = g(\mathbf{x}, t) \quad \text{for all } \mathbf{x} \in \partial\Omega \text{ and all } t.$$

where  $\hat{n}$  is the unit vector (pointing to the outside) normal to the boundary at each point  $\mathbf{x} \in \partial\Omega$ . In other words, it is the unit vector orthogonal to the tangent space of  $\partial\Omega$  at each point.

Examples:

- heat flux,
- normal electric field.

### C. Robin (Mixed)

$$a u(\mathbf{x}, t) + b \frac{\partial u}{\partial \hat{n}}(\mathbf{x}, t) = g(\mathbf{x}, t) \quad \text{for all } \mathbf{x} \in \partial\Omega \text{ and all } t.$$

Examples:

- convective heat transfer.

### D. Periodic

$$u(\mathbf{x} + L, t) = u(\mathbf{x}, t).$$

Examples:

- rings,
- crystals,
- Fourier analysis.

## V. SEPARATION OF VARIABLES

Assume

$$u(\mathbf{x}, t) = X(\mathbf{x})T(t).$$

Then

$$\frac{\mathcal{M}T}{T} = \frac{\mathcal{L}X}{X} = -\lambda.$$

This produces:

- A **spatial problem**

$$\mathcal{L}X = -\lambda X.$$

- and a **temporal problem** determined by the PDE under study.

Thus every separable PDE ultimately reduces to solving the a few spatial eigenvalue problems, since the heat, wave, Schrödinger, Poisson, etc., all involve the same spatial operator  $\mathcal{L}$ , and therefore the same eigenfunctions  $\phi_n$ .

## VI. COORDINATE SYSTEMS AND EIGENFUNCTIONS

### A. Cartesian Coordinates

#### 1. Interval $[0, L]$

Eigenproblem:

$$X'' + \lambda X = 0.$$

- Dirichlet

$$X(0) = X(L) = 0.$$

Eigenfunctions:

$$X_n(x) = \sin \frac{n\pi x}{L}.$$

Eigenvalues:

$$\lambda_n = \left( \frac{n\pi}{L} \right)^2.$$

- Neumann

$$X'(0) = X'(L) = 0.$$

Eigenfunctions:

$$X_n(x) = \cos \frac{n\pi x}{L}.$$

Eigenvalues:

$$\lambda_n = \left( \frac{n\pi}{L} \right)^2.$$

## 2. Periodic

$$X(0) = X(L), \quad X'(0) = X'(L).$$

Eigenfunctions:

$$e^{i2\pi nx/L}.$$

Eigenvalues:

$$\lambda_n = \left( \frac{2\pi n}{L} \right)^2.$$

## B. Rectangular Domains

$$0 < x < L_x, \quad 0 < y < L_y.$$

Dirichlet eigenfunctions:

$$\phi_{nm}(x, y) = \sin \frac{n\pi x}{L_x} \sin \frac{m\pi y}{L_y}.$$

Eigenvalues:

$$\lambda_{nm} = \left( \frac{n\pi}{L_x} \right)^2 + \left( \frac{m\pi}{L_y} \right)^2.$$

### C. Polar Coordinates

$$(r, \theta).$$

Laplacian:

$$\nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}.$$

- Angular Part

$$\Theta'' + m^2 \Theta = 0.$$

Eigenfunctions:

$$e^{im\theta}.$$

Eigenvalues:

$$m^2.$$

- Radial Part

$$r^2 R'' + r R' + (k^2 r^2 - m^2) R = 0.$$

Solutions:

$$J_m(kr), \quad Y_m(kr).$$

(Bessel functions)

### D. Disk of Radius $a$

Regular eigenfunctions:

$$J_m(k_{mn}r)e^{im\theta}.$$

with

$$J_m(k_{mn}a) = 0.$$

Eigenvalues:

$$\lambda_{mn} = k_{mn}^2 \left( \frac{j_{mn}}{a} \right)^2.$$

where  $j_{mn}$  denotes the  $n$ -th positive zero of  $J_m$ .

### E. Cylindrical Coordinates

$$(r, \phi, z).$$

Eigenfunctions:

$$J_m(k_r r) e^{im\phi} e^{ik_z z}.$$

Eigenvalues:

$$\lambda = k_r^2 + k_z^2.$$

where the dependence of  $\lambda$  on  $m$  goes through the value  $k_r$  that appears in the radial equation.

## F. Spherical Coordinates

$$(r, \theta, \phi).$$

- Angular Part

$$L^2 Y = \ell(\ell + 1)Y.$$

Eigenfunctions:

$$Y_\ell^m(\theta, \phi).$$

(Spherical harmonics) Eigenvalues:

$$\ell(\ell + 1).$$

- Radial Part

$$r^2 R'' + 2rR' + (k^2 r^2 - \ell(\ell + 1)) R = 0.$$

The solutions are:

$$j_\ell(kr) \quad y_\ell(kr)$$

where

$$j_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+\frac{1}{2}}(x) \quad y_\ell(x) = \sqrt{\frac{\pi}{2x}} Y_{\ell+\frac{1}{2}}(x)$$

are the spherical Bessel functions.

- Sphere of Radius  $a$

Eigenfunctions:

$$j_\ell(k_{\ell n} r) Y_\ell^m(\theta, \phi).$$

with

$$j_\ell(k_{\ell n} a) = 0.$$

Eigenvalues:

$$\lambda_{\ell n} = k_{\ell n}^2.$$

## VII. COMMON SPECIAL FUNCTIONS

## VIII. SOLVING PDES USING THE EIGENBASIS

Once

$$\mathcal{L}\phi_n = -\lambda_n \phi_n$$

is known, expand

$$u = \sum_n a_n(t) \phi_n.$$

The PDE reduces to equations for the coefficients  $a_n(t)$ .

TABLE I: Special Functions and Differential Equations

| Function                           | Differential Equation                       |
|------------------------------------|---------------------------------------------|
| Bessel ( $J_\nu, Y_\nu$ )          | $x^2 y'' + xy' + (x^2 - \nu^2)y = 0$        |
| Legendre ( $P_\ell$ )              | $(1 - x^2)y'' - 2xy' + \ell(\ell + 1)y = 0$ |
| Associated Legendre ( $P_\ell^m$ ) | Angular equation in spherical coordinates   |
| Hermite ( $H_n$ )                  | Harmonic oscillator                         |
| Laguerre ( $L_n^\alpha$ )          | Hydrogen atom radial equation               |

## IX. LAPLACE EQUATION

$$\nabla^2 u = 0.$$

No source term. Solutions are harmonic functions. Usually solved directly by separation of variables.

## X. POISSON EQUATION

$$\nabla^2 u = f.$$

Expand

$$f = \sum_n f_n \phi_n.$$

Then

$$-\lambda_n a_n = f_n.$$

Therefore

$$a_n = -\frac{f_n}{\lambda_n}.$$

Hence

$$u = -\sum_n \frac{f_n}{\lambda_n} \phi_n.$$

## XI. HELMHOLTZ EQUATION

$$\nabla^2 u + k^2 u = 0.$$

Equivalent to

$$\mathcal{L}u = -k^2 u.$$

This is precisely the spatial eigenvalue problem.

## XII. HEAT EQUATION

$$u_t = D\nabla^2 u.$$

Expand

$$u = \sum_n a_n(t) \phi_n.$$

Then

$$a'_n = -D\lambda_n a_n.$$

Solution:

$$a_n(t) = a_n(0) e^{-D\lambda_n t}.$$

Therefore

$$u(\mathbf{x}, t) = \sum_n a_n(0) e^{-D\lambda_n t} \phi_n(\mathbf{x}).$$

## XIII. WAVE EQUATION

$$u_{tt} = c^2 \nabla^2 u.$$

Expansion gives

$$a''_n + c^2 \lambda_n a_n = 0.$$

Solution:

$$a_n(t) = A_n \cos(c\sqrt{\lambda_n}t) + B_n \sin(c\sqrt{\lambda_n}t).$$

Hence

$$u = \sum_n \left[ A_n \cos(c\sqrt{\lambda_n}t) + B_n \sin(c\sqrt{\lambda_n}t) \right] \phi_n.$$

## XIV. SCHRÖDINGER EQUATION

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi.$$

Expansion yields

$$a'_n = -i \frac{\hbar \lambda_n}{2m} a_n.$$

Thus

$$a_n(t) = a_n(0) e^{-iE_n t/\hbar},$$

where

$$E_n = \frac{\hbar^2 \lambda_n}{2m}.$$



## XV. GREEN FUNCTIONS

For

$$\mathcal{L}u = f,$$

define

$$\mathcal{L}G(x, x') = \delta(x - x').$$

Then

$$u(x) = \int G(x, x') f(x') dx'.$$

### A. Spectral Representation

If

$$\mathcal{L}\phi_n = -\lambda_n \phi_n,$$

then

$$G(x, x') = - \sum_n \frac{\phi_n(x) \phi_n(x')}{\lambda_n}.$$

Thus Green functions are simply the inverse operator written in the eigenbasis.

**Observation:** Previous expansion assumes that all eigenvalues satisfy  $\lambda_n \neq 0$ . If a zero mode exists (e.g. for Neumann or periodic boundary conditions), then the operator  $\mathcal{L}$  is not invertible and the Green function must be modified by excluding the zero mode:

$$G(x, x') = - \sum_{\lambda_n \neq 0} \frac{\phi_n(x) \phi_n(x')}{\lambda_n}.$$

In this case, solutions exist only if the source  $f$  is orthogonal to the kernel of  $\mathcal{L}$  (compatibility condition).

[TODO: Talk about the particular case in which the Green function can be written using the Wronksian.]

## XVI. FOURIER METHODS

Fourier methods are particularly useful on infinite or periodic domains.

### A. Fourier Series

If  $u(x)$  is periodic with period  $L$ ,

$$u(x + L) = u(x),$$

it can be expanded as

$$u(x) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi n x/L},$$

where

$$c_n = \frac{1}{L} \int_0^L u(x) e^{-i2\pi nx/L} dx.$$

The functions

$$e^{i2\pi nx/L}$$

are eigenfunctions of the derivative operator:

$$\frac{d}{dx} e^{i2\pi nx/L} = i \frac{2\pi n}{L} e^{i2\pi nx/L}.$$

Consequently,

$$\frac{d}{dx} \longrightarrow i \frac{2\pi n}{L},$$

and

$$\frac{d^2}{dx^2} \longrightarrow - \left( \frac{2\pi n}{L} \right)^2.$$

Thus constant-coefficient differential equations become algebraic equations for the Fourier coefficients  $c_n$ .

## B. Fourier Transform

For problems on the infinite line,

$$-\infty < x < \infty,$$

the discrete Fourier series is replaced by the Fourier transform

$$\hat{u}(k) = \int_{-\infty}^{\infty} u(x) e^{-ikx} dx,$$

with inverse transform

$$u(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{u}(k) e^{ikx} dk.$$

The Fourier transform may be regarded as the continuum limit of the Fourier series as  $L \rightarrow \infty$ . Differentiation becomes multiplication:

$$\frac{d}{dx} \longrightarrow ik,$$

$$\frac{d^2}{dx^2} \longrightarrow -k^2.$$

More generally,

$$\frac{d^n}{dx^n} \longrightarrow (ik)^n.$$

Conversely, multiplication by  $x$  becomes differentiation in Fourier space:

$$x \longrightarrow i \frac{d}{dk},$$

$$x^n \longrightarrow i^n \frac{d^n}{dk^n}.$$

Therefore constant-coefficient differential equations become algebraic equations in Fourier space, while variable-coefficient equations generally remain differential equations.

### Typical Applications:

- Heat equation on  $\mathbb{R}$ .
- Wave equation on  $\mathbb{R}$ .
- Poisson equation on  $\mathbb{R}^n$ .
- Construction of Green functions.
- Spectral analysis of translation-invariant operators.

## XVII. MASTER FORMULA

Almost every linear PDE solved in a standard Mathematical Methods course eventually takes the form

$$u(\mathbf{x}, t) = \sum_n a_n(t) \phi_n(\mathbf{x}),$$

where:

- $\phi_n$  are eigenfunctions of the spatial operator,
- $\lambda_n$  are the corresponding eigenvalues,
- $a_n(t)$  are determined by the temporal equation,
- coefficients are fixed by initial and boundary conditions.

The entire theory can therefore be summarized as

|                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{PDE} \longrightarrow \text{Eigenvalue Problem} \longrightarrow \text{Eigenfunctions} \longrightarrow \text{Series Expansion} \longrightarrow \text{Solution}$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

which unifies Fourier series, Bessel functions, spherical harmonics, Sturm-Liouville theory, Green functions, and separation of variables into a single spectral framework.

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